

Hardware, Software and Firmware — Detailed Lecture Notes

JKSSB Computer Awareness

October 8, 2026

17.1 Hardware

Hardware is the set of physical, tangible parts of a computer system — the parts you can see and touch. A computer's hardware includes the central processing unit (CPU), the motherboard, random access memory (RAM), the hard disk drive or solid-state drive, the keyboard, the mouse, the monitor, the printer, and every other physical component. Hardware carries out the actual work of the machine: it executes instructions, stores data, and moves signals.

Hardware alone does nothing useful. A processor with no instructions to run is only silicon and metal. The instructions come from software, and the two together make a working computer. For exam purposes, the test for hardware is simple: if it is a physical object that you can touch, it is hardware.

Definition (Hardware). *The physical equipment of a computer system — the tangible components such as the CPU, memory, storage, input devices, output devices, and communication hardware.*

Hardware is usually grouped by function. Input devices, such as the keyboard, mouse, and scanner, bring data into the computer. Output devices, such as the monitor, printer, and speaker, present results. Processing hardware, mainly the CPU and the graphics processing unit (GPU), performs computation. Memory hardware, such as RAM, ROM, and cache, holds data and instructions. Storage hardware, such as the hard disk drive (HDD) and solid-state drive (SSD), keeps files without power. Communication hardware, such as the network interface card and the motherboard, connects the parts together.

Hardware is also described as internal or external. **Internal hardware** is fixed inside the computer case — the motherboard, CPU, RAM, and internal drives. **External hardware**, also called a peripheral, is connected from outside, such as the keyboard, mouse, monitor, printer, and an external drive. A question may ask whether a named part is internal or external; the touch test still applies, because both kinds are physical and both are hardware.

A single concrete machine runs through this course: the **office desktop** — a personal computer with a CPU, memory, a disk, a keyboard, a mouse, a monitor, and a printer, running an operating system and MS Word. Every term in this lesson can be pointed at on that one machine, which makes the abstract layers easier to remember.

17.2 Software

Software is the set of programs and instructions that tell the hardware what to do. Unlike hardware, software is intangible: you cannot touch it. It is kept on secondary storage, such as a hard disk or solid-state drive, and it is loaded into main memory when it runs. Software is what turns a collection of physical parts into a usable computer.

Definition (Software). *The collection of programs, procedures, and related data that direct the operation of a computer. Software is logical and intangible, and it instructs the hardware.*

Software falls into two broad kinds. **System software** runs and manages the computer itself — the operating system, device drivers, and utility programs. **Application software** helps the user perform a task — a word processor such as MS Word, a web browser, a spreadsheet, or a game. Every program a candidate meets in JKSSB questions belongs to one of these two kinds, but the boundary matters: the operating system is system software, not application software.

Software is kept on secondary storage as files. When it is launched, the operating system copies it into main memory, and the CPU executes it from there. This is why a program must be *loaded* before it can run, and why data held only in RAM is lost when the power goes off. The permanent copy lives on disk; the running copy lives in memory.

17.3 Hardware vs software

The two are easy to confuse only when the wording is careless, so hold the contrasts in a table.

Point	Hardware	Software
Nature	Physical and tangible	Logical and intangible
Can be touched?	Yes	No
Role	Executes the work	Gives the instructions
Stored or fixed	Fixed as physical parts	Stored as files, loaded to run
Example	CPU, printer, keyboard	MS Word, a driver, an OS
How it fails	Wears out, breaks, burns out	Bugs, corruption, wrong logic

Neither can work alone. Hardware and software together form a computer system, and a question that names one and describes the other is usually testing this single distinction. A candidate should be able to sort a mixed list — a CPU, MS Word, a printer driver, the BIOS, and a user — into its correct class at a glance.

Example 17.3.1. *A question lists “CPU, MS Word, printer driver, BIOS, operator” and asks which are software. **MS Word** and the **printer driver** are software. The CPU is hardware; the BIOS is firmware (software stored in hardware); and the operator is a person, that is, humanware. Sorting is done by asking, for each item, whether it is physical, a program, embedded code, or a person.*

17.4 System software and application software

System software manages the machine and provides services to other programs. The operating system is the main example, together with device drivers and system utilities. **Application software** is written to let the user accomplish a task, and it depends on the system software beneath it.

For the exam, the stable rule is: the operating system is system software. This is the point tested by the real Junior Assistant 2026 concept recorded as PYQ-06 (device driver = software; OS = system software). A device driver, too, is system software — it is a program, not a physical part.

17.5 Firmware

Firmware is software that is embedded in hardware and stored in non-volatile memory. It is held in read-only memory (ROM), erasable programmable ROM (EPROM), or flash memory inside the device, so it is available the moment the device is powered on. Firmware provides the low-level control instructions a device needs and starts it up.

Definition (Firmware). *Software permanently stored in a hardware device, usually in non-volatile memory such as ROM or flash. It is software by nature but is treated as part of the hardware because it is fixed to a specific device.*

Common examples of firmware include the **BIOS** (Basic Input Output System) or **UEFI** on a personal computer, which initialises the hardware and begins loading the operating system; the firmware inside a printer, a router, a keyboard, or a solid-state drive. Firmware is usually described as sitting between hardware and software. It is a program, so it is software; it is stored on the physical device, so it behaves like part of the hardware.

Firmware differs from ordinary software in how tightly it is bound to the device. Firmware is written for one specific piece of hardware, is stored in ROM or flash on that device, and changes rarely (it is updated by a process called flashing). Ordinary software is general-purpose, is stored on secondary storage, and is installed and updated frequently.

Firmware is the **first software to run** when a device is switched on. On a personal computer, the CPU executes the BIOS or UEFI firmware from ROM before any operating system is available; that firmware checks the hardware and then hands control to the operating system. This is exactly why firmware must be non-volatile: it has to be present even before anything can be loaded. A candidate should not confuse firmware with the operating system — the firmware runs first and *starts* the computer, while the operating system is what the machine is finally running.

17.6 Device drivers

A device driver is a small program that lets the operating system communicate with a piece of hardware. It acts as a translator: the operating system sends a general request such as “print this page”, and the driver converts it into the exact signals that a specific printer understands. A printer driver, for example, lets the operating system print on one particular printer model.

Definition (Device driver). *A program that enables the operating system to communicate with, and control, a specific hardware device by translating general commands into device-specific signals.*

The single most important exam point about a driver is that it is **software**, even though it is tied to hardware and is often named after a device. A candidate may be tempted to call a driver hardware because it belongs to a printer or a keyboard, but that is wrong. The driver is a program, it is stored like software, and a device will not work properly without the right driver installed. This is TRAP-10, and it is also the heart of the real Junior Assistant 2026 concept PYQ-06.

Example 17.6.1. *A question states, “A device driver is a hardware component that connects the printer to the computer.” This is wrong on two counts: the driver is a **program**, not a physical component, and it is software, not hardware. The physical cable and port are hardware; the driver that makes the printer work is software. The trap is that the driver carries the device’s name.*

17.7 Humanware

Humanware, also called **peopleware**, is the human component of a computer system. It covers everyone involved: the end users who operate the machine, the programmers who write software, the system administrators who maintain it, and the operators who run it. A computer system is usually described as hardware, software, and humanware together.

Definition (Humanware). *The people who design, operate, maintain, and use a computer system; the human component that hardware and software ultimately serve.*

Humanware is not a device and not a program. In a classification question, if the named item is a person or a role — user, programmer, operator, administrator — it is humanware. It is the layer that makes decisions and uses the results, and without it the hardware and software would have no purpose.

17.8 The layered computing stack

The parts of a computer system can be arranged as a stack of layers, each one resting on the layer below. The order, from the bottom up, is:

Layer	Role
Hardware	The physical components at the base of the system
Firmware	Stored on the hardware; starts it and controls it at a low level
Device driver	Lets the operating system communicate with the hardware
Operating system	Manages all resources and provides services to programs
Application software	Lets the user perform a task

Reading the stack from the bottom, the physical parts come first. Firmware sits directly on them and brings the hardware to life. The device driver sits above the firmware and gives the operating system a clean way to talk to each device. The operating system manages memory, files, processes, and devices, and offers services to the programs above it. The application software at the top is what the user actually opens and works in. A command travels *down* the stack — from an application, through the operating system and the driver, to the hardware — and results travel back up.

This stack is the single mental picture for the whole lesson: hardware at the bottom, software above it, and firmware and drivers as the thin bridges that connect the two.

17.9 Why the stack matters

The layered stack is not only a way to sort terms; it explains the order in which work is done. When a user clicks Print in a word processor, the command begins at the application-software layer. The application asks the operating system to print; the operating system passes the job to the device driver for that printer; the driver translates the request into the signals the printer hardware understands; and the printer’s own firmware controls the physical printing. The result then travels back up the stack to the user. Every layer depends on the one below it, and a fault in any single layer can stop the whole chain — which is why “reinstall the driver” is such a common fix.

Example 17.9.1. *A printer fails to print although the paper and ink are fine. Because the system has layers, the fault could lie in the application (a wrong page range), the operating system (a stuck print queue), the driver (missing or outdated), or the hardware (a loose cable). Testing from the bottom up checks each layer in turn. In exam terms, the layer that “lets the operating system talk to the printer” is the **device driver**.*

17.10 Exam retrieval and common confusions

Five distinctions prevent most errors on this topic.

1. Hardware is **physical and tangible**; software is **logical and intangible**. The simple test is touch: if you can touch it, it is hardware.
2. A **device driver** is **software**, not hardware. Its name follows the device, but it is a program (TRAP-10; PYQ-06).
3. The **operating system** is **system software**, not application software (PYQ-06).
4. **Firmware** is software stored in non-volatile memory inside hardware, such as the BIOS; it is software by nature, hardware by storage.
5. **Humanware** (peopleware) is the **people** in the system, not a device or a program.

Example 17.10.1. *An item asks, “Which of the following is software?” with options CPU, a printer, a device driver, and a keyboard. The answer is the **device driver**: it is a program, whereas the CPU, the printer, and the keyboard are physical. If the options instead are an operating system, MS Word, a BIOS chip, and a game, the item is testing system software versus application software and firmware, and the operating system is the system software.*

Exercises

1. Define hardware and software, and give one example of each.
2. Distinguish system software from application software. To which kind does the operating system belong?
3. What is firmware, where is it stored, and why is it called “between hardware and software”?
4. Explain why a device driver is classified as software even though it is named after a hardware device.
5. What is humanware? Give two roles it includes.
6. List the layers of the computing stack from the bottom (hardware) to the top (application).
7. Give one example each of internal hardware and external hardware.
8. Using the print command as an example, explain how a request travels down the computing stack.

Solutions

1. Hardware is the physical, tangible equipment of a computer, such as the CPU or a printer. Software is the programs and instructions that tell the hardware what to do, such as MS Word or a device driver.
2. System software runs and manages the computer itself — the operating system, device drivers, and utilities. Application software lets the user do a task, such as a word processor or a game. The operating system is **system software**.
3. Firmware is software embedded in a hardware device and stored in non-volatile memory such as ROM or flash, for example the BIOS. It is software by nature but is fixed to a specific device, so it is described as sitting between hardware and software.
4. A device driver is a program, so it is software. It is named after the device it serves, which tempts a candidate to call it hardware, but it is stored and works like other software; it merely translates the operating system's commands for one specific device.
5. Humanware, or peopleware, is the human component of a computer system. It includes roles such as end users and system administrators (also programmers or operators).
6. Hardware, then firmware, then device driver, then operating system, then application software.
7. Internal hardware: the motherboard (also the CPU, RAM, or an internal drive). External hardware: the keyboard (also the mouse, monitor, printer, or an external drive).
8. The user clicks Print in the application software. The application asks the operating system, which passes the job to the printer's device driver. The driver translates the command into the signals the printer hardware understands, and the printer's firmware controls the physical printing. The result returns up the stack to the user.